DRAFT

MINIMUM DEGREE EIGHT IN K_7^--MINOR-FREE CONTRACTION-CRITICAL GRAPHS

CAVIT ERGINSOY

Abstract. Let G be a minor-minimal non-six-colourable graph with no K_7^- minor. For each i, let n_i count the vertices of degree i, and put $\tau = \sum_{i\geq 10}(i-9)n_i$. We prove that G is seven-connected, has no vertices of degree seven, contains no K_5 subgraph, and satisfies

$$|E(G)| \geq 4|V(G)| \qquad \text{and} \qquad n_8 \geq 25 + \tau.$$

In particular, $\delta(G) \geq 8$, and the neighbourhood of every degree-eight vertex is K_4-free. A classification of degree-seven neighbourhoods, followed by a rooted $K_{4,2}^*$ argument, gives the exclusion. We also prove that every $(r+1)$-connected graph containing two distinct K_r subgraphs has a K_{r+2}^- minor. The proof is computation-free.

1. Introduction

Hadwiger's conjecture states that every graph of chromatic number t contains a K_t minor [3]. It is known for $t \leq 6$, with the last case due to Robertson, Seymour and Thomas [9], and remains open for $t \geq 7$. Write K_t^- for the graph obtained from K_t by deleting one edge, and $K_7^{\vee}$ for the graph obtained from K_7 by deleting two incident edges. Albar proved that every K_7^--minor-free graph is seven-colourable [1, Theorem 1]. Norin and Totschnig proved that every $K_7^{\vee}$-minor-free graph is six-colourable [8, Theorem 4]. They asked whether K_7^- may replace $K_7^{\vee}$ in that statement [8, Conjecture 21].

A graph is *minor-minimal non-six-colourable* if it is not six-colourable and every proper minor is six-colourable. The following theorem describes such a graph under the exclusion in Conjecture 21.

Theorem 1.1. *Let G be a minor-minimal non-six-colourable K_7^--minor-free graph. For each i, let n_i denote the number of vertices of degree i, and put*

$$\tau = \sum_{i\geq 10}(i-9)n_i.$$

Date: Draft, 11 August 2026.

2020 *Mathematics Subject Classification.* 05C15, 05C40.

Key words and phrases. Hadwiger's conjecture, graph minors, contraction-critical graphs, rooted minors.

Then G is seven-connected and

$$n_7 = 0, \qquad \delta(G) \geq 8, \qquad |E(G)| \geq 4|V(G)|.$$

Moreover, G contains no K_5 subgraph,

$$n_8 \geq 25 + \tau,$$

and the neighbourhood of every degree-eight vertex is K_4-free.

The theorem does not settle Conjecture 21. It reduces the conjecture to the open assertion that every seven-connected graph H satisfying $|E(H)| \geq 4|V(H)|$ contains a K_7^- minor.

Mayer obtained a finite catalogue of candidate neighbourhood graphs for degree-seven vertices in six-contraction-critical graphs [7]. In the minor-minimal $K_7^{\vee}$-minor-free setting, Norin and Totschnig proved that every degree-seven vertex belongs to a K_5 contained in its closed neighbourhood [8, Claim 4.4]. We first determine the whole neighbourhood in the K_7^- setting, then rule out degree seven.

Rolek and Song proved the next result when $r = 6$ [10, Lemma 1.9]. We shall use the uniform form with $r = 5$.

Theorem 1.2. *For every $r \geq 2$, if G is an $(r+1)$-connected graph containing two distinct K_r subgraphs, then $K_{r+2}^- \preccurlyeq G$.*

The connectivity hypothesis is sharp. The graph K_{r+1} is r-connected and contains two distinct K_r subgraphs, but has no K_{r+2}^- minor.

The following rooted closure will exclude the remaining degree-seven neighbourhood.

Theorem 1.3. *Let G be a six-connected graph of order n, let $v \in V(G)$ have degree $d \geq 5$, and suppose that $G[N(v)]$ contains a K_4 subgraph. If*

$$|E(G)| \geq 4n + d - 13,$$

then $K_7^- \preccurlyeq G$.

A *minor model* of a graph H in G is a family $(B_x : x \in V(H))$ of pairwise disjoint connected vertex sets such that B_x and B_y are adjacent whenever $xy \in E(H)$. A model is *rooted* at nominated vertices when its branch sets contain them one-to-one. We write $H \preccurlyeq G$ when H is a minor of G. All graphs are finite and simple.

2. Two linked cliques

Lemma 2.1. *Let $r \geq 2$. If an $(r+1)$-connected graph H contains two vertex-disjoint K_r subgraphs, then $K_{r+2}^- \preccurlyeq H$. The branch sets can be chosen to meet the union of the two cliques.*

Proof. Write the two cliques as

$$A = \{a_1, \ldots, a_r\}, \qquad B = \{b_1, \ldots, b_r\}.$$

By Menger's theorem there are r pairwise vertex-disjoint A to B paths whose open interiors avoid $A \cup B$. Relabel them as paths P_i from a_i to b_i, and put $U = \bigcup_i V(P_i)$.

Call a path Q a *usable cross-path* if its ends lie on distinct paths P_i, P_j, its open interior avoids U, and its ends are not both A-ends or both B-ends. Suppose first that Q exists. Split each of P_i and P_j across an edge into a nonempty A-prefix and a nonempty B-suffix, choosing the split so that the ends of Q belong to pieces of opposite kinds. Absorb the open interior of Q into the piece containing one end. The four resulting pieces are connected and disjoint. The two A-pieces are adjacent through A, the two B-pieces are adjacent through B, the two pieces from each original path are adjacent across its splitting edge, and Q supplies one of the two remaining cross-adjacencies. Thus at most one adjacency among the four pieces is missing. Each unsplit path is adjacent to every split piece through its two clique ends, and the unsplit paths are pairwise adjacent. These $r+2$ sets form a K_{r+2}^- model.

Suppose now that no usable cross-path exists. Let C be a component of $H-U$. If C has neighbours on two distinct paths, a path through C between two such attachments is unusable only when both attachments are A-ends or both are B-ends. Comparing all pairs of attachments shows that all neighbours of C are A-ends or all are B-ends. Hence $|N_H(C)| \le r$, contrary to $(r+1)$-connectivity. Every component of $H-U$ therefore attaches to only one P_i.

For each i, let W_i be the union of the open interior of P_i and all components of $H-U$ that attach to P_i. No vertex of W_i has a neighbour on P_j for $j \ne i$. Indeed, an edge from the open interior of P_i to P_j would be a usable cross-path, while such an edge from a component of $H-U$ would make that component attach to two distinct linkage paths, contrary to the preceding paragraph. Hence

$$N_H(W_i) \subseteq \{a_i, b_i\}.$$

Since $r \ge 2$, a nonempty W_i would give a vertex cut of order at most two. Thus every W_i is empty. It follows that H consists of A, B and the matching edges $a_i b_i$: any other edge between A and B would be usable. This graph has minimum degree r, again contradicting $(r+1)$-connectivity. □

Proof of Theorem 1.2. Let L_1, L_2 be distinct K_r subgraphs and put $s = |V(L_1) \cap V(L_2)|$. If $s \le r-2$, delete the common set Z. The graph $G-Z$ is $(r+1-s)$-connected and contains the two disjoint K_{r-s} subgraphs $L_1 - Z$ and $L_2 - Z$. Lemma 2.1 gives a K_{r-s+2}^- model in which every branch set contains a vertex of $(L_1 \cup L_2) - Z$. The vertices of Z, taken as singleton branch sets, are complete to this model and to one another. They complete a K_{r+2}^- model.

It remains that $s = r-1$. The union $X = V(L_1) \cup V(L_2)$ induces a graph containing K_{r+1}^-. Since an $(r+1)$-connected graph has at least $r+2$ vertices, $G-X$ is nonempty. Let C be a component of $G-X$. If C missed a vertex of X, then $N_G(C)$ would be a cut of order at most r. Thus C is adjacent to every vertex of X. The vertices of X as singleton branch sets, together with C, form a K_{r+2}^- model. $\square$

We shall use Theorem 1.2 with $r = 5$.

Corollary 2.2. *Every six-connected K_7^--minor-free graph contains at most one K_5 subgraph.*

3. Degree-seven neighbourhoods

A graph is *k-contraction-critical* if it has chromatic number k and every proper minor is $(k-1)$-colourable. We use two standard facts. A noncomplete k-contraction-critical graph is seven-connected when $k \geq 7$ [6, Satz 6]; see also [10, Theorem 1.8]. If G is k-contraction-critical and $v \in V(G)$, then

$$\alpha(G[N(v)]) \leq d(v) - k + 2. \tag{3.1}$$

This inequality goes back to Dirac [2]; we use the formulation in [10, Lemma 1.6(i)].

We also use the following specialisation of Theorem 7 and property $(*)$ of Kriesell and Mohr [5, Theorem 7]. Let a graph J be properly coloured with five colours, and choose one root from each colour class. Let D be a graph on the five roots with at most six edges. If the ends of every edge $xy \in E(D)$ lie in the same bichromatic component of J on their two colours, then J contains pairwise disjoint connected sets, one containing each root, that are adjacent for every edge of D.

Lemma 3.1. *Let G be a seven-chromatic graph such that every proper minor is six-colourable and $K_7^- \not\preccurlyeq G$. Let v have degree seven, put $H = G[N(v)]$, and let ab be a nonedge of H. If*

$$R = N(v) \setminus \{a, b\},$$

then $G - v - \{a, b\}$ contains a K_5 minor rooted at R.

Proof. Contract the two edges va and vb to one vertex. A six-colouring of the resulting proper minor induces a six-colouring φ of $G - v$ in which a and b have one common colour, say 0. Every vertex of R avoids colour 0. The five vertices of R use the other five colours distinctly, since a colour absent from $N(v)$ could be assigned to v.

Put $F = \overline{H}$. Inequality (3.1) gives $\alpha(H) \leq 2$, so F is triangle-free. If $xy \in E(F[R])$ and x, y have colours i, j, then x, y lie in the same component of the subgraph of $G - v$ induced by colours i, j. Otherwise, interchanging i and j on the component containing x would remove colour i from $N(v)$, allowing that colour at v.

Delete the colour-0 class from $G - v$. The resulting graph is properly five-coloured, with R a transversal of its colour classes. By Mantel's theorem, the triangle-free graph $F[R]$ has at most six edges. Apply the Kriesell and Mohr result with demand graph $F[R]$. It gives five disjoint connected sets rooted at R and adjacent for every nonedge of $H[R]$. Every other pair is joined by its edge in $H[R]$, so the five sets form the required rooted K_5 model. $\square$

Theorem 3.2. *Under the hypotheses of Lemma 3.1, every degree-seven vertex v satisfies*

$$G[N(v)] \cong K_4 \mathbin{\dot\cup} K_3 \quad \text{or} \quad K_1 \vee (K_3 \mathbin{\dot\cup} K_3). \tag{3.2}$$

In particular, v belongs to a K_5 subgraph.

Proof. Put $H = G[N(v)]$ and $F = \overline{H}$. By (3.1), F is triangle-free. It is nonempty, since otherwise $G[N[v]]$ would contain K_7^-.

Every nonisolated vertex of F has degree at least three. To see this, choose an edge $ab \in E(F)$ and use the rooted model of Lemma 3.1 on $R = N(v) \setminus \{a, b\}$. If $d_F(a) = 1$, then a is adjacent in H to every root; the rooted model, $\{a\}$ and $\{v\}$ form a K_7 model. If $d_F(a) = 2$, with other neighbour x, then triangle-freeness gives $bx \in E(H)$. Absorb b into the branch set rooted at x. Together with $\{a\}$ and $\{v\}$, the resulting seven branch sets have at most one missing adjacency, again a contradiction.

Every nontrivial component of F therefore has minimum degree at least three and hence at least six vertices. Since $|V(F)| = 7$, there is one nontrivial component and at most one isolated vertex. If the component has six vertices, Mantel's theorem and its equality case give $F \cong K_{3,3} \mathbin{\dot\cup} K_1$.

Suppose instead that the component has seven vertices. No vertex has degree at least five: its independent neighbourhood would contain a vertex of degree at most two. Thus every degree is three or four. The degree sum is even, so some vertex z has degree four. Its four neighbours are independent. Each is adjacent to both vertices outside $N_F[z]$ to have degree at least three, and those two outside vertices are nonadjacent by triangle-freeness. Hence $F \cong K_{3,4}$. Taking complements gives (3.2). $\square$

Corollary 3.3. *Let G be a minor-minimal non-six-colourable K_7^--minor-free graph. Then G contains at most one K_5 subgraph and every degree-seven vertex v satisfies*

$$G[N(v)] \cong K_4 \mathbin{\dot\cup} K_3.$$

Proof. Albar's theorem gives $\chi(G) = 7$, while minor minimality makes every proper minor six-colourable. Corollary 2.2 gives at most one K_5 subgraph. The second graph in (3.2) would place v in two distinct K_5 subgraphs, so only the first can occur. $\square$

4. Rooted $K_{4,2}^*$ closure

For a graph F and $S \subseteq V(F)$, say that (F, S) is *internally k-connected* if there is no separation (A, B) of F with $S \subseteq A$, $B \setminus A \neq \varnothing$, and $|A \cap B| < k$. A Z-rooted $K_{4,2}^*$ model consists of four pairwise disjoint connected branch sets, one containing each member of the four-set Z, and two further branch sets U, V. Each of U, V is adjacent to all four rooted branch sets and to the other; no adjacency between distinct rooted branch sets is required.

We use the following bound of Norin and Totschnig [8, Lemma 12].

Theorem 4.1 (Norin and Totschnig)**.** *Let F be a graph and let $Z \subseteq V(F)$ have order four. If (F, Z) is internally four-connected and F has no Z-rooted $K^*_{4,2}$ model, then*

$$|E(F)| \leq 4|V(F)| - 10. \tag{4.1}$$

The following placement lemma puts one prescribed vertex in a non-root branch set.

Lemma 4.2. *Let F be a graph and let $S = Z \dot\cup \{x\}$, where $|Z| = 4$. Suppose that (F, S) is internally five-connected. If F has a Z-rooted $K^*_{4,2}$ model, then it has one in which x belongs to a non-root branch set.*

Proof. Choose a Z-rooted model that maximises $|U| + |V|$ over its two non-root branch sets U, V and, subject to that, minimises $\sum_{i=1}^{4} |R_i|$ over its rooted branch sets $R_1, \dots, R_4$. Write z_i for the root in R_i, and put

$$P_i = \{r \in R_i : r \text{ has a neighbour in } U \cup V\}.$$

We claim that $|P_i| = 1$. Both U and V meet R_i, so P_i is nonempty. If it had at least two vertices, there would be distinct $u, w \in R_i$ such that u meets U and w meets V; otherwise the two contact sets would be the same singleton. Take a minimal tree in $F[R_i]$ containing z_i, u, w. By the minimal choice of the rooted branch sets, this tree spans R_i. One of u, w, say u, is a leaf different from z_i. Move u from R_i into U. Both altered branch sets remain connected. The former tree edge from u to $R_i \setminus \{u\}$ preserves the contact between R_i and U, while w preserves the contact between R_i and V. Every other required adjacency is unchanged, contrary to the choice of the model. Hence $|P_i| = 1$.

No component outside the six branch sets can meet $U \cup V$, since it could be absorbed into whichever of U, V it meets. It follows that

$$|N_F(U \cup V) \setminus (U \cup V)| \leq 4, \tag{4.2}$$

with at most one external neighbour in each rooted branch set. If $x \notin U \cup V$, (4.2) would give a separation of (F, S) of order at most four whose second open side is the nonempty set $U \cup V$. This contradicts internal five-connectivity. Thus $x \in U \cup V$. $\square$

Proof of Theorem 1.3. Choose a four-set $Z \subseteq N_G(v)$ inducing K_4, choose $x \in N_G(v) \setminus Z$, and put $F = G - v$. Deleting one vertex from a six-connected graph leaves a five-connected graph. In particular, (F, Z) is internally four-connected and $(F, Z \cup \{x\})$ is internally five-connected. Moreover,

$$\begin{aligned} |E(F)| &= |E(G)| - d \\ &\geq 4n - 13 = 4|V(F)| - 9. \end{aligned} \tag{4.3}$$

Theorem 4.1 supplies a Z-rooted $K^*_{4,2}$ model in F, and Lemma 4.2 lets us choose it with x in one non-root branch set.

The four rooted branch sets are pairwise adjacent through the edges of $G[Z] = K_4$. Together with the two non-root branch sets they form a K_6 model. The singleton branch set $\{v\}$ is adjacent to all four rooted branch sets and to the one containing x; it may miss only the other. These seven branch sets form a K_7^- model in G. □

5. Consequences for a critical graph

Jakobsen proved that an n-vertex graph with $n \geq 7$ and at least $\frac{9}{2}n - 12$ edges contains a K_7^- minor or is a $(K_{2,2,2,2}, K_6, 4)$-cockade [4, Theorem 4]; see also [1, Theorem 2].

Lemma 5.1. *Let G be a minor-minimal non-six-colourable K_7^--minor-free graph, with $n = |V(G)|$ and $m = |E(G)|$. Then*

$$2m \leq 9n - 25. \tag{5.1}$$

Proof. The graph G is neither of the two base graphs of the cockade family, since both K_6 and $K_{2,2,2,2}$ are six-colourable. A nontrivial cockade has a vertex cut of order four, whereas G is seven-connected. Thus G is not an exceptional cockade. Since it has no K_7^- minor, Jakobsen's theorem gives $m < \frac{9}{2}n - 12$. Integrality yields (5.1). □

Proposition 5.2. *Let G be a minor-minimal non-six-colourable K_7^--minor-free graph. Then*

$$n_7 \leq 5, \qquad |E(G)| \geq 4|V(G)| - 2. \tag{5.2}$$

Proof. By Corollary 3.3, every degree-seven vertex lies in the unique possible K_5, so $n_7 \leq 5$. Seven-connectivity gives minimum degree at least seven, and every vertex not counted by n_7 has degree at least eight. Hence

$$2|E(G)| \geq 7n_7 + 8(|V(G)| - n_7) = 8|V(G)| - n_7 \geq 8|V(G)| - 5.$$

The left side is even, so it is at least $8|V(G)| - 4$, proving (5.2). □

Proof of Theorem 1.1. Albar's theorem gives $\chi(G) = 7$, and every proper minor is six-colourable by the choice of G. The graph is noncomplete because it has no K_7^- minor, so Mader's theorem gives seven-connectivity.

Suppose that v has degree seven. Corollary 3.3 shows that $G[N(v)]$ contains a K_4. Proposition 5.2 gives

$$|E(G)| \geq 4|V(G)| - 2 > 4|V(G)| - 6,$$

which exceeds the threshold in Theorem 1.3 when $d_G(v) = 7$. That theorem gives the forbidden K_7^- minor. Thus $n_7 = 0$. Seven-connectivity now gives $\delta(G) \geq 8$, and degree summation gives $|E(G)| \geq 4|V(G)|$.

Put $q = |E(G)| - 4|V(G)|$, so $q \geq 0$. Suppose that a K_5 subgraph has vertex set K. For each $w \in K$, the neighbourhood of w contains the K_4 on $K \setminus \{w\}$. Since G has no K_7^- minor, the contrapositive of Theorem 1.3 gives

$$4|V(G)| + q < 4|V(G)| + d_G(w) - 13,$$

and hence $d_G(w) \geq q + 14$. The five vertices of K would contribute at least $5(q + 6)$ to

$$\sum_{z \in V(G)} (d_G(z) - 8) = 2q,$$

which is impossible. Thus G contains no K_5 subgraph.

Finally, Lemma 5.1 and the exact degree identity give

$$25 \leq 9|V(G)| - 2|E(G)| = n_8 - \tau.$$

Therefore $n_8 \geq 25+\tau$. If a degree-eight vertex had a K_4 in its neighbourhood, it would belong to a K_5, which has just been excluded. □

Statement on AI use and independence

GPT-5.6 Sol, an OpenAI model, contributed to theorem discovery, proofs, verification and drafting. Cavit Erginsoy directed the work and made the final decisions. Erginsoy is not affiliated with OpenAI, and OpenAI neither sponsored nor verified the work.

References

1. Boris Albar, *Coloration of K_7^--minor free graphs*, 2014, arXiv:1402.2806.
2. G. A. Dirac, *Trennende Knotenpunktmengen und Reduzibilität abstrakter Graphen mit Anwendung auf das Vierfarbenproblem*, Journal für die reine und angewandte Mathematik **204** (1960), 116-131.
3. Hugo Hadwiger, *Über eine Klassifikation der Streckenkomplexe*, Vierteljahrsschrift der Naturforschenden Gesellschaft in Zürich **88** (1943), 133-142.
4. Ivan Tafteberg Jakobsen, *On a certain homomorphism properties of graphs II*, Mathematica Scandinavica **52** (1983), 229-261.
5. Matthias Kriesell and Samuel Mohr, *Kempe chains and rooted minors*, 2019, arXiv:1911.09998, version 2, revised 2022.
6. Wolfgang Mader, *Über trennende Eckenmengen in homomorphiekritischen Graphen*, Mathematische Annalen **175** (1968), 243-252.
7. Jean Mayer, *Case 6 of Hadwiger's Conjecture. III. The Problem of 7-Vertices*, Discrete Mathematics **111** (1993), no. 1-3, 381-387.
8. Sergey Norin and Agnès Totschnig, *Every graph with no $K_7^\vee$-minor is 6-colorable*, 2025, arXiv:2507.03244.
9. Neil Robertson, Paul Seymour, and Robin Thomas, *Hadwiger's conjecture for K_6-free graphs*, Combinatorica **13** (1993), no. 3, 279-361.
10. Martin Rolek and Zi-Xia Song, *Coloring graphs with forbidden minors*, Journal of Combinatorial Theory, Series B **127** (2017), 14-31.